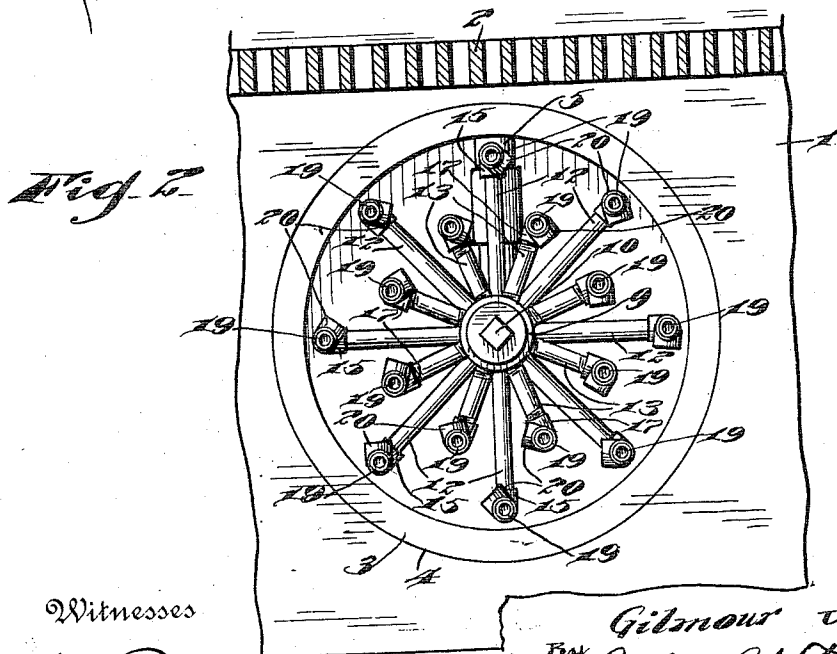
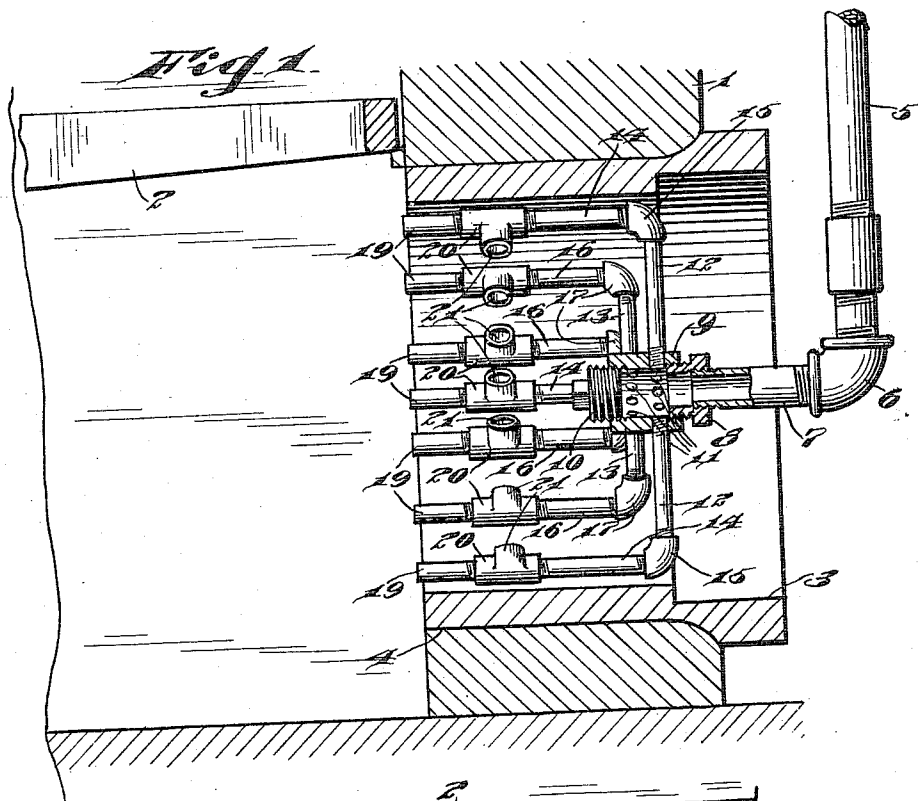


985,415.

G. JOHN.
FORCE DRAFT APPARATUS.
APPLICATION FILED JAN. 19, 1910.

Patented Feb. 28, 1911.

2 SHEETS—SHEET 1.



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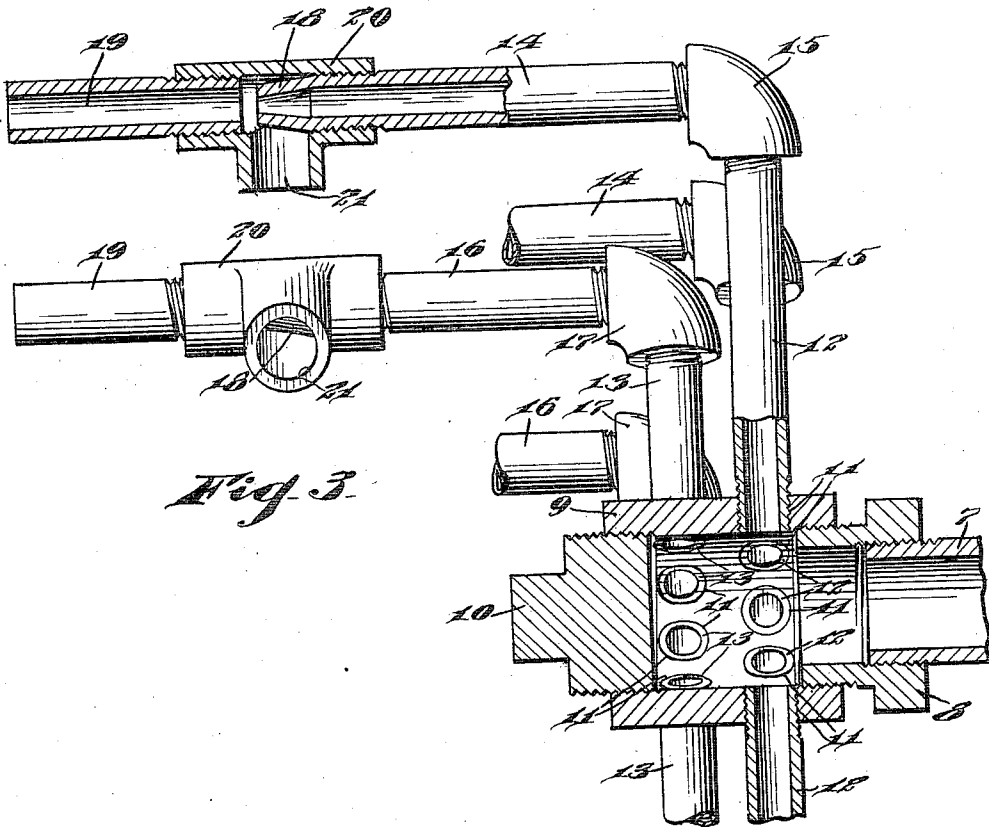


Fig. 3.

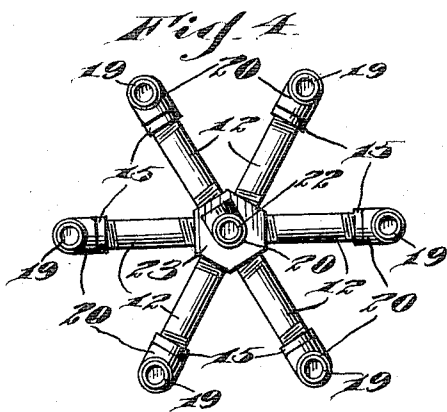


Fig. 4.

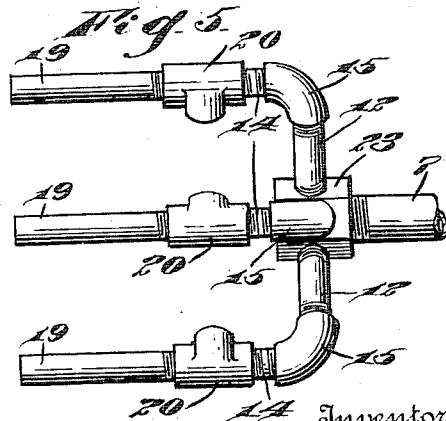


Fig. 5.

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FORCE-DRAFT APPARATUS.

985,415.

Specification of Letters Patent.

Patented Feb. 28, 1911.

Application filed January 19, 1910. Serial No. 533,916.

To all whom it may concern:

Be it known that I, GILMOUR JOHN, a citizen of the United States, residing at Shamokin, in the county of Northumberland and State of Pennsylvania, have invented certain new and useful Improvements in Force-Draft Apparatus, of which the following is a specification.

My invention relates to improvements in force draft apparatus, the object of the invention being to provide a construction of steam discharge pipes or nozzles through which air is drawn by the steam and forced into the furnace below the grate.

A further object is to provide an improved construction of steam outlet or nozzle and improved mounting therefor, whereby a circular series of steam ejectors are provided with air inlets to each, so that a maximum of air is carried into the furnace with the steam and an economy in steam is the result, as well as an extremely strong draft.

With these and other objects in view, the invention consists in certain novel features of construction and combinations and arrangements of parts, as will be more fully hereinafter described and pointed out in the claim.

In the accompanying drawings: Figure 1 is a view in section through a furnace wall illustrating my improvements. Fig. 2 is a view in section through the furnace at right angles to Fig. 1, showing the force draft apparatus in end elevation. Fig. 3 is an enlarged fragmentary view, partly in elevation and partly in section of the force draft apparatus, and Figs. 4 and 5 are views in end and side elevation respectively, illustrating a modification.

1 represents a furnace wall and 2 the furnace grate.

3 represents a pipe which may be made of terra cotta or other suitable material, which is preferably inserted in an opening 4 in wall 1, and in this pipe 3 is located my improved force draft apparatus, which will now be described in detail.

5 represents a steam supply pipe, which is connected with any suitable source of steam supply, and is connected at its lower end by an elbow 6 with a horizontal pipe 7, which latter projects into the center of pipe 3. This pipe 7 is connected by an internal and externally screw threaded coupling 8, with an internally screw threaded collar 9, the end of which is closed by a threaded

plug 10. This collar 9 is provided with two circular series of screw threaded openings 11, said openings of one series being disposed preferably midway between the openings of the other series, and two circular series of radial pipes 12 and 13 are screwed into these openings 11. It will be noted that the radial pipes 12 are appreciably longer than pipes 13, so that the series of horizontally disposed pipes 14 at the ends of pipes 12 and connected thereto by elbows 15, are disposed an appreciable distance radially, farther away from collar 9 than are pipes 16, which are connected by elbows 17 with pipes 13 but are located in a circle concentric therewith.

The pipes 14 and 16 are made at their ends in the form of contracted nozzles 18 to secure the greatest kinetic energy of the steam as it discharges, and this discharge from the nozzles 18 is had through pipes 19 in line with nozzles 18, and yet spaced therefrom by means of T-couplings 20, which are screw threaded at their ends to receive the nozzles and the pipes 19, and also provide air entrance nipples 21 parallel across which the nozzles 18 project, so that the kinetic energy developed by the steam in its exit through the nozzles 18 will draw the air through the entrance 21 and discharge it with the steam through the pipes 19 below the grate 2, to secure the maximum draft and the maximum proportion of air with the steam, which is the design of practically most force draft apparatus.

It will be noted that as pipes 12 and 13 are not in alinement, and as pipes 13 are shorter than pipes 12, the greatest possible area of the outer portion of tube 3 is covered by the pipes 14 and 16, so that each air inlet 21 is given as large an air space to draw its air from as is possible, and air will be drawn into the tube 1 from the outside and enter the several air inlets and be discharged into the furnace.

While I do not, of course, limit myself to any one particular disposition of the air inlet, one desirable form is illustrated in Fig. 2 in which the air inlets are all disposed at an angle to the pipes 12 and 13, and this has been found to be a very effective formation.

In the modification shown in Figs. 4 and 5, but a single circular series of ejector pipes 19 are provided, but in addition, a single ejector pipe 22 is provided at the center

and in line with the pipe 7. All of these pipes are connected to a hexagonal block 23 coupled to pipe 7.

Various slight changes might be made in the general form and arrangement of parts described without departing from my invention, and hence I do not restrict myself to the precise details set forth, but consider myself at liberty to make such changes and alterations as fairly fall within the spirit and scope of the appended claim.

Having thus described my invention, what I claim as new and desire to secure by Letters Patent is:—

15 A force draft apparatus, comprising a steam supply pipe, a collar, an internally and externally screw-threaded coupling connecting said collar and steam pipe, said collar having two annular series of screw-
20 threaded openings, a plug normally closing the end of said collar, corresponding series

of longer and shorter radial pipes screwed into said openings in the collar, the shorter pipes alternating with the longer, elbows on the ends of said radial pipes, T-couplings, the intermediate nipple of each T-coupling projecting inwardly, two concentric annular series of horizontal pipes screwed into said elbows and into said T-couplings, and having restricted nozzles at their ends inside of the T-couplings, and discharge pipes screwed into the T-couplings in line with the nozzles, substantially as described.

In testimony whereof I have signed my name to this specification in the presence of two subscribing witnesses.

GILMOUR JOHN.

Witnesses:

JESSE JOHN,
A. J. CAMPBELL.

Copies of this patent may be obtained for five cents each, by addressing the "Commissioner of Patents, Washington, D. C."